FRE-GY-9743 Assignment 6: A GBP-Collateralised EUR/USD Forward, and Swap Products

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Code: github.com/pawarraj8888/FRE-GY-9743-Assignments-6, branch `Raj_Pawar_Branch`.
Project page: pawarraj8888.github.io/FRE-GY-9743-Assignments-6.

Part I: an EUR/USD forward collateralised in GBP

Setting. At T I receive N EUR and pay NK USD, so K is in USD per EUR, like the spot rate X_t. The trade sits under a CSA with cash collateral in GBP, remunerated at SONIA, and I assume perfect collateralisation as in the notes. I write c_{USD}, c_{EUR}, c_{GBP} for the collateral (OIS) rates SOFR, EuroSTR and SONIA, $r_{\text{USD}}, r_{\text{EUR}}, r_{\text{GBP}}$ for the rate at which a desk borrows and invests cash in each currency (one rate per currency, which collapses the notes' r_R and r_F; the gap between those two is the FVA question and I leave it aside), and $y_i := r_i - c_i$ for the gap between the cash rate and the collateral rate in currency i. The y_i are small and not directly observable; they only ever appear as differences, which is what the cross-currency basis market quotes.

Discounting a payoff whose collateral is in another currency. I redo the replication argument of the collateral notes (Section 2.2) for a USD payoff V collateralised in GBP. The counterparty posts collateral worth V, i.e. $C_{\text{GBP}} = V/X_t^{\text{GBPUSD}}$ pounds, on which I pay SONIA, $c_{\text{GBP}}C_{\text{GBP}}$. I keep the pounds in GBP and invest them at r_{GBP} (converting them would leave me short pounds against a liability that is marked in pounds), so the collateral account nets me $(r_{\text{GBP}} - c_{\text{GBP}})C_{\text{GBP}}$ per unit time, which is $y_{\text{GBP}}V$ in dollars at spot. The derivative itself I fund by borrowing V dollars at r_{USD}, and the hedge is financed at the secured rate exactly as in Eq. (12) of the notes. Collecting the carry terms, the self-financing condition becomes $\partial_t V + \mathcal{L}V = r_{\text{USD}}V - y_{\text{GBP}}V$, and Feynman–Kac gives

$$V_t = \mathbb{E}_t^{Q_{\text{USD}}}\Big[e^{-\int_t^T (r_{\text{USD}} - y_{\text{GBP}})\,du}\, V_T\Big] = \mathbb{E}_t^{Q_{\text{USD}}}\Big[e^{-\int_t^T (c_{\text{USD}} + y_{\text{USD}} - y_{\text{GBP}})\,du}\, V_T\Big].$$

Two checks: with GBP collateral on a GBP payoff the pounds repay the borrowing directly, the round trip disappears and the rate is $r_{\text{GBP}} - y_{\text{GBP}} = c_{\text{GBP}}$, SONIA discounting, which is Eq. (16) of the notes; with USD collateral on a USD payoff it is SOFR. So the general rule is: a payoff in currency i collateralised in currency k is discounted at $c_i + y_i - y_k$, and I write the corresponding collateralised zero-coupon bond as

$$P_i^{(k)}(t,T) := \mathbb{E}_t^{Q_i}\Big[e^{-\int_t^T (c_i + y_i - y_k)\,du}\Big], \qquad P_i^{(i)}(t,T) = \mathbb{E}_t^{Q_i}\Big[e^{-\int_t^T c_i\,du}\Big] \text{ the OIS discount factor.}$$

Q1: valuation formula. The USD leg is a fixed payment of NK at T under GBP collateral, worth $-NK\,P_{\text{USD}}^{(\text{GBP})}(t,T)$. The EUR leg is N euros at T under the same CSA; the argument above run in EUR values it at $N\,P_{\text{EUR}}^{(\text{GBP})}(t,T)$ euros, and a euro value today is X_t dollars (a spot trade, no funding involved). Hence

$$V_t = N\Big[X_t\,P_{\text{EUR}}^{(\text{GBP})}(t,T) - K\,P_{\text{USD}}^{(\text{GBP})}(t,T)\Big] = N\,P_{\text{USD}}^{(\text{GBP})}(t,T)\big[F^{(\text{GBP})}(t,T) - K\big], \qquad F^{(\text{GBP})}(t,T) := X_t\,\frac{P_{\text{EUR}}^{(\text{GBP})}(t,T)}{P_{\text{USD}}^{(\text{GBP})}(t,T)}.$$

In the language of Section 2.3 of the notes, $P_{\text{USD}}^{(\text{GBP})}(t,T)$ is the numeraire of the T-forward measure for this CSA, X_T is a martingale under it with $F^{(\text{GBP})}(t,T) = \mathbb{E}_t^{T,\text{GBP}}[X_T]$, and $V_t = N\,P_{\text{USD}}^{(\text{GBP})}(t,T)\,\mathbb{E}_t^{T,\text{GBP}}[X_T - K]$.

Q2: fair forward strike. $V_t = 0$ gives

$$K^* = F^{(\text{GBP})}(t,T) = X_t\,\frac{P_{\text{EUR}}^{(\text{GBP})}(t,T)}{P_{\text{USD}}^{(\text{GBP})}(t,T)},$$

covered interest parity with both legs discounted on their GBP-collateralised curves. If the spreads y are deterministic the expectation factorises, $P_i^{(k)} = P_i^{(i)}\,e^{-\int_t^T (y_i - y_k)du}$, and the collateral currency drops out of the ratio: $K^* = X_t\,(P_{\text{EUR}}^{(\text{EUR})}/P_{\text{USD}}^{(\text{USD})})\,e^{-\int_t^T (y_{\text{EUR}} - y_{\text{USD}})du}$, the same forward whatever the CSA. The collateral currency changes how the mark-to-market is discounted, not where the forward is.

Q3: are the five discount factors enough? First what they are. $\text{df}_{\text{SONIA}} = P_{\text{GBP}}^{(\text{GBP})}$, $\text{df}_{\text{EuroSTR}} = P_{\text{EUR}}^{(\text{EUR})}$ and $\text{df}_{\text{SOFR}} = P_{\text{USD}}^{(\text{USD})}$ are the OIS curves, cashflows collateralised in their own currency. $\text{df}_{[\text{GBP,USD}]} = P_{\text{GBP}}^{(\text{USD})}$ and $\text{df}_{[\text{EUR,USD}]} = P_{\text{EUR}}^{(\text{USD})}$ are GBP and EUR cashflows collateralised in USD, which is what GBP/USD and EUR/USD cross-currency basis swaps under a USD CSA bootstrap.[1] What I need are $P_{\text{USD}}^{(\text{GBP})}$ and $P_{\text{EUR}}^{(\text{GBP})}$, and neither is on the list, so taken literally the answer is no. But every $P_i^{(k)}$ differs from the OIS curve $P_i^{(i)}$ only through the collateral switch $e^{-\int (y_i - y_k)}$, and the y's enter only as differences. Under the assumption that any curve construction already makes when it turns these expectations into discount factors, that the spreads are deterministic (or at least independent

[1] If the bracket is meant the other way round, USD cashflows collateralised in GBP and in EUR, the same algebra goes through with the roles swapped: $P_{\text{USD}}^{(\text{GBP})} = \text{df}_{[\text{GBP,USD}]}$ directly and $P_{\text{EUR}}^{(\text{GBP})} = \text{df}_{\text{EuroSTR}}\,\text{df}_{[\text{GBP,USD}]}/\text{df}_{[\text{EUR,USD}]}$. The conclusion does not change.

of the rates and of FX), the switch factors chain: with $Y_{i,k}(T) := e^{-\int_t^T (y_i - y_k)du}$ I have $Y_{i,k} = Y_{i,\mathrm{USD}}\, Y_{\mathrm{USD},k}$ and $Y_{k,i} = 1/Y_{i,k}$, and the two USD bases give $Y_{\mathrm{GBP,USD}} = \mathrm{df}_{[\mathrm{GBP,USD}]}/\mathrm{df}_{\mathrm{SONIA}}$ and $Y_{\mathrm{EUR,USD}} = \mathrm{df}_{[\mathrm{EUR,USD}]}/\mathrm{df}_{\mathrm{EuroSTR}}$. Then

$$P^{(\mathrm{GBP})}_{\mathrm{USD}} = \mathrm{df}_{\mathrm{SOFR}}\, Y_{\mathrm{USD,GBP}} = \mathrm{df}_{\mathrm{SOFR}}\, \frac{\mathrm{df}_{\mathrm{SONIA}}}{\mathrm{df}_{[\mathrm{GBP,USD}]}}, \qquad P^{(\mathrm{GBP})}_{\mathrm{EUR}} = \mathrm{df}_{\mathrm{EuroSTR}}\, Y_{\mathrm{EUR,USD}}\, Y_{\mathrm{USD,GBP}} = \mathrm{df}_{[\mathrm{EUR,USD}]}\, \frac{\mathrm{df}_{\mathrm{SONIA}}}{\mathrm{df}_{[\mathrm{GBP,USD}]}},$$

and the forward and its value become

$$K^* = X_t\, \frac{\mathrm{df}_{[\mathrm{EUR,USD}]}(T)}{\mathrm{df}_{\mathrm{SOFR}}(T)}, \qquad V_t = N\, \frac{\mathrm{df}_{\mathrm{SONIA}}(T)}{\mathrm{df}_{[\mathrm{GBP,USD}]}(T)} \Big[X_t\, \mathrm{df}_{[\mathrm{EUR,USD}]}(T) - K\, \mathrm{df}_{\mathrm{SOFR}}(T) \Big].$$

So yes, they are sufficient, and the way GBP enters is instructive. The fair strike needs only the USD-collateralised EUR curve and SOFR: it is the ordinary USD-CSA forward, as Q2 said it had to be. The GBP collateral appears only in the factor $\mathrm{df}_{\mathrm{SONIA}}/\mathrm{df}_{[\mathrm{GBP,USD}]} = P^{(\mathrm{GBP})}_{\mathrm{USD}}/P^{(\mathrm{USD})}_{\mathrm{USD}}$, the ratio of the GBP-collateralised to the USD-collateralised USD discount factor, which rescales the mark-to-market $K^* - K$. Equivalently, the EUR/GBP collateral basis is triangulated from the two USD bases. Three caveats I would attach to that answer. The chaining uses the deterministic-spread assumption; if the bases are stochastic and correlated with FX, each $P_i^{(k)}$ carries a convexity term that the five curves do not determine. The triangulated EUR/GBP basis is a no-arbitrage value; where EUR/GBP basis swaps trade at a different level, a GBP-collateralised EUR curve bootstrapped from them should replace the triangulation, and the formula above becomes the check against it. And if a desk can invest collateral at something other than the market cash rate, the y's become desk-specific: that is the cheapest-to-deliver collateral question, not a curve question.

Part II: swap products in the valuation library

I completed the six `TODO` blocks following the supplied FRA example. In `linear_products.py`, `ProductRFRSwap` builds a fixed leg and a compounded-overnight leg with `InterestRateStream`; both legs run from the adjusted effective date to the termination date, their accrual schedules follow the overnight index's fixing calendar with the swap's payment business-day convention, and both pay with the swap's payment offset and conventions. The floating leg carries the swap's compounding method and spread, which `ProductOvernightIndexCashflow` adds outside the daily compounding. `ProductOvernightIndexBasisSwap` builds two such overnight legs through one small helper, each on its own index, calendar and accrual period, with the spread on leg 1 only. The supplied index-only and spread-only views of leg 1 reuse the leg-1 schedule, so leg 1 values to exactly their sum. In `product_factory.py` the two builders read everything but the dates and the quote from the convention; the shared date rule treats `1Y` as spot-starting, `3Mx1Y` as a 3M forward start running for 1Y, and accepts explicit dates in either slot; the quote is the fixed rate for the swap and the leg-1 spread for the basis swap. In `valuation_engine.py` the two `calculate_value` methods are signed sums of the supplied leg engines' values and of their cash.

Numbers. Value date 2026-01-02, notional $N = 1,000,000$ USD, trades from 2026-04-02 to 2027-04-02, funding on SOFR plus the IFR spread. A useful identity for checking the floating legs by hand: when every observation period of the index calendar lies inside the coupon, the daily factors $D(s_j)/D(e_j)$ telescope, so a compounded coupon pays $N\big(D(T_s)/D(T_e) - 1\big)$ on the projection curve plus $N\alpha s$ for the spread, discounted with the funding curve at the payment date. For the annual SOFR coupon this gives 4.2730%, as in the notebook's cashflow table.

Product	Leg values (USD)	PV (USD)	Fair level
FRA 3×6, receive 4% (the supplied example)	fixed 9,937.00, floating 9,849.91	87.09	
RFR swap, receive 4% vs SOFR, 1Y/1Y	fixed 38,534.54, floating 41,164.22	-2,629.68	par rate 4.2730%
Basis swap, SOFR + 10 bp vs Fed Funds, 3M/3M	leg 1 42,163.00 (index 41,182.75, spread 980.25), leg 2 33,716.53	8,446.47	fair spread -76.17 bp

Start	End	Pays	α	SOFR cmpd %	FF cmpd %	D_{fund}	Leg 1 PV	Leg 2 PV
2026-04-02	2026-07-02	2026-07-07	0.252778	3.9649	2.7713	0.982228	10,092.66	6,880.75
2026-07-02	2026-10-02	2026-10-06	0.255556	3.9652	3.6664	0.972239	10,100.34	9,109.54
2026-10-02	2027-01-04	2027-01-06	0.261111	3.9656	3.6668	0.962191	10,214.35	9,212.31
2027-01-04	2027-04-02	2027-04-06	0.244444	4.9614	3.6656	0.950167	11,755.64	8,513.93

The swap is negative for the fixed receiver because 4% is below the 4.27% that SOFR compounds to; the basis swap is worth 8,446.47 because Fed Funds projects 1 to 1.3 points below SOFR on these synthetic curves, far more than the 10 bp spread. Every number in the notebook agrees with the hand calculation above, and the 41 tests in `tests/` check the same things: leg construction and dates, the factory's rules, the engines against the hand numbers, par rate and fair spread, arithmetic compounding, same-day cash from fixings and past payments, the reports and the registry.